Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 36: The Same Numbers, Everywhere

Pascal's Triangle and the Binomial Theorem

Chartrand & Zhang, Section 9.1, pp. 313–322

Name: _____ Date: _____

Where we are

The numbers $\binom{n}{r}$ turn up in three places that look unrelated: counting subsets, a triangle built by adding neighbours, and the coefficients when you expand $(x+y)^n$.
Today we see they are the same numbers, and why.

By the end of today you should be able to

1. Build Pascal's triangle and state the identity behind it.
2. Prove that identity by a counting argument.
3. Use the Binomial Theorem to find a specified coefficient.

1. The triangle

Each entry is the sum of the two directly above it. Fill in rows 5 and 6.

row 0				1			
row 1			1		1		
row 2		1		2		1	
row 3	1		3		3		1
row 4	1	4	6	4	1		
row 5	1	5	10	10	5	1	
row 6	1	6	15	20	15	6	1

Row n consists of $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$ — the number of subsets of each size. So row 4 is $1, 4, 6, 4, 1$, matching $\binom{4}{0}$ through $\binom{4}{4}$.

Two patterns you can already explain. Each row is symmetric, because $\binom{n}{r} = \binom{n}{n-r}$. And row n sums to 2^n, because it counts all subsets by size.

2. Why neighbours add

Theorem 9.1 — Pascal's identity

For $1 \le r \le n-1$: $$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}.$$

Proof by counting one thing two ways. Let $S = \{a_1, \dots, a_n\}$. The left side counts all r-element subsets of S. Now split those subsets into two cases according to whether they contain a_n.

Containing a_n: the other $r-1$ elements come from the remaining $n-1$, so there are $\binom{n-1}{r-1}$.

Not containing a_n: all r elements come from the remaining $n-1$, so there are $\binom{n-1}{r}$.

The two cases are disjoint and cover everything, so by the Addition Principle their counts sum to $\binom{n}{r}$. ■

A combinatorial proof

No algebra with factorials — just a case split on one element. Proofs like this are called **combinatorial proofs**: count a single collection in two ways and equate the results.
The case split is Lecture 24's technique; the addition is Lecture 31's principle.

Check it. $\binom{6}{2} = \binom{5}{1} + \binom{5}{2}$, that is, $15 = 5 + 10$. ✓

3. The Binomial Theorem

Expand by hand and compare with the triangle:

$$(x+y)^2 = x^2 + 2xy + y^2, \qquad (x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3.$$

The coefficients are rows 2 and 3.

Theorem 9.4 — the Binomial Theorem

For every nonnegative integer n,

$$(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^{n-r} y^r.$$

Why. Expanding $(x+y)^n = (x+y)(x+y)\cdots(x+y)$ means choosing x or y from each of the n factors. A term $x^{n-r}y^r$ arises by choosing y from exactly r of the factors — and the number of ways to do that is $\binom{n}{r}$.

So the coefficients are counts. That is the whole content of the theorem.

4. Extracting a single coefficient

Usually you want one coefficient, not the whole expansion.

Find the coefficient of x^3y^2 in $(x+y)^5$. Here $n = 5$ and $r = 2$, giving $\binom{5}{2} = 10$.

Find the coefficient of x^4y^3 in $(2x+y)^7$. The general term is

$$\binom{7}{r}(2x)^{7-r}y^r = \binom{7}{r}2^{7-r}x^{7-r}y^r.$$

We need $7 - r = 4$, so $r = 3$, and the coefficient is $\binom{7}{3} \cdot 2^4 = 35 \cdot 16 = 560$.

Do not forget the constant

With $(2x + y)^n$ the 2 is raised to a power too. Writing just $\binom{7}{3} = 35$ misses a factor of 16. **Always write the general term first**, then solve for the r you need. Guessing which binomial coefficient to use is how these go wrong.

Your turn. Find the constant term in $\left(x + \frac{1}{x}\right)^6$.

General term: $\binom{6}{r} x^{6-r} x^{-r} = \binom{6}{r} x^{6-2r}$. Constant means $6 - 2r = 0$, so $r = 3$, giving 20.

Exit ticket

1. Write row 7 of Pascal's triangle.

2. Find the coefficient of $x^2 y^4$ in $(x + y)^6$.

Before Monday

Read the Chapter 8 Highlights, pp. 308–309.

Self-check (not collected). Section 9.1, Exercises 5, 7, and 9 — odd-numbered, answers in the back.

Coming up

Homework 12 is due tonight, Friday, November 13, 11:59 p.m.
Quiz 11 is Monday, November 16 — Section 8.4.
Homework 13 covers Sections 8.4, 8.5, and 9.1 and is due Friday, November 20.
Graphs begin Monday — Chapter 12, the last new unit.

Questions I still have